ANALYSIS OF BLU-DROP (Limitations & Future Enhancement)

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***Abstract*—Peer-to-peer (P2P) communication systems using Bluetooth and Wi-Fi Direct enable decentralized data exchange without infrastructure. However, flooding-based communication suffers from severe scalability limitations, including broadcast storms, excessive memory consumption, and high latency due to redundant message propagation. This paper presents a comprehensive analysis of these limitations and proposes a hierarchical clustering-based routing model combined with Bloom filter-based memory optimization. Analytical modeling and numerical evaluation demonstrate that the proposed approach reduces the transmission overhead from $3N + 1$ to $N/4 + 1$, while reducing hop count from $O(N)$ to $O(\log N)$, and memory usage by up to 60×. The results validate the effectiveness of structured communication and probabilistic data structures for large-scale decentralized networks.**

***Index Terms*—P2P, Bluetooth, Wi-Fi Direct, Bloom Filter, Hierarchical Clustering, Scalability.**

I. INTRODUCTION

Peer-to-peer communication plays a critical role in applications such as disaster response, proximity-based sharing, and decentralized networking. Technologies such as Bluetooth Low Energy (BLE) and Wi-Fi Direct, integrated through APIs like Google Nearby Connections, enable direct communication between devices without centralized infrastructure.

Despite their advantages, these systems are primarily designed for small-scale deployments. As network size increases, flooding-based communication results in redundant message propagation, leading to broadcast storms, increased latency, high energy consumption, and excessive memory usage.

This paper addresses these challenges by:

- Analyzing limitations of flooding-based communication
- Proposing hierarchical clustering with routing optimization
- Introducing Bloom filter-based memory optimization
- Providing numerical and analytical validation of scalability improvements

II. SYSTEM MODEL & METHODOLOGY

II-A. Network Assumptions

- Total number of devices in the network = N
- Average number of neighbouring nodes per node = $k \approx 4$ (based on P2P_CLUSTER constraints)
- Cluster size: 4 devices (1 Cluster Head (CH), 3 Member Devices)
- Number of clusters = $C = N/4$

II-B. Performance Metrics

- Total Transmissions (T)
- Hop Count (H)
- Memory Consumption (M)

II-C. Evaluation Methodology

- Analytical modeling of communication and memory complexity
- Numerical evaluation using scalable dataset
- Comparative analysis between flooding and proposed models

III. EXISTING MODEL: FLOODING ALGORITHM

In the flooding-based model, each node rebroadcasts received messages to all its neighbors. While duplicate suppression prevents reprocessing, it does not eliminate redundant transmissions, resulting in excessive network overhead.

III-A. Communication Complexity Approximation

1) Fully Connected Graph

In the worst-case scenario of a fully connected network:

1) Every device broadcast message to every neighbour device.
2) N (Devices) $\times N$ (Transmission).
3) P2P Cluster strategy provide 3–5 stable connections.
4) Due to redundancy and overlapping of path the complexity remains $O(N^2)$.

$$\boxed{T_{flood} = O(N^2)} \tag{1}$$

2) Real-life Connection Graph

To evaluate the communication overhead of the current BluDrop system, the network is modeled as an undirected graph $G(V, E)$, where V represents the set of devices and E represents the set of connections. For the experimental topology (see Fig. 1), we assume $N = 10$ and $E = 11$.

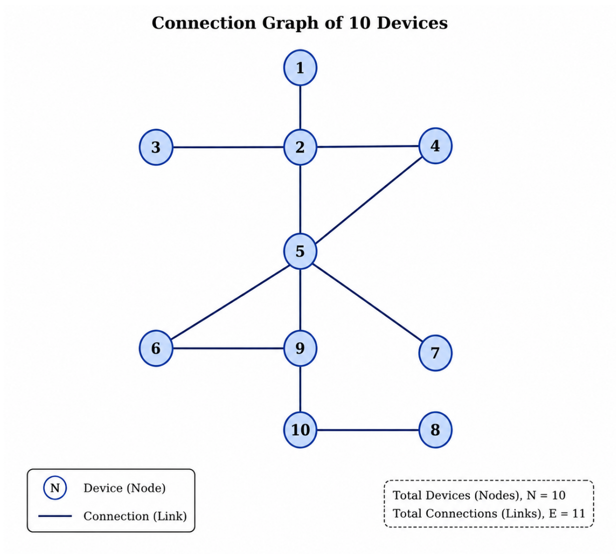

Fig. 1. Connection Graph of 10 Devices (Sample Case Study).

Controlled Flooding Model: In the implemented protocol, each device forwards a received message to all connected neighbors except the sender. Duplicate messages are suppressed using unique message identifiers. Thus, each node processes and forwards a message at most once.

Transmission Derivation: In the worst case, each edge may carry the message in both directions, giving $T_{max} = 2E$. However, reverse transmissions along the $N - 1$ edges of the spanning tree are avoided because the sender endpoint is excluded during forwarding. Therefore, the effective number of transmissions is $T = 2E - (N - 1)$. For a graph with average degree ≈ 4, $E \approx 2N$, resulting in:

$$\boxed{T_{existing} = 2(2N) - (N - 1) = 3N + 1} \tag{2}$$

The complexity remains $O(N)$, but with a significant constant factor.

III-B. Memory Complexity Approximation

To guarantee duplicate suppression and prevent infinite message cycling in the unstructured flooding graph, each node must keep a historical cache of all processed message identifiers.

In the existing architecture, every message is trackable via a standard Universally Unique Identifier (UUID) represented as a string token.

1) A canonical string-encoded UUID requires a baseline footprint of 36 bytes for character storage.
2) In typical high-level memory managed environments (such as the JVM or runtime Android ART environment), storing these strings inside standard look-up sets (e.g., `HashSet` or `LinkedHashSet`) adds significant object header, reference pointer, and bucket allocation padding overhead.
3) This inflates the practical memory cost to approximately 72 bytes per unique message reference.

For an incoming batch stream containing m total messages, the memory consumption pattern scales strictly linearly with respect to the transaction volume:

$$\boxed{M_{existing} = m \times S_{uuid}} \tag{3}$$

Where $S_{uuid} \approx 72$ bytes. As shown by this model, a continuous or high-density throughput creates an un-bounded linear accumulation footprint ($O(m)$ space complexity). This rapidly eats up volatile heap storage on resource-constrained peripheral edge mobile devices.

III-C. Hop Count Complexity Approximation

In the current flooding-based system, the hop count grows approximately linearly with the number of devices:

$$\boxed{H_{existing} = O(N)} \tag{4}$$

IV. PROPOSED MODEL: HIERARCHICAL CLUSTERING WITH TARGETED ROUTING AND BLOOM FILTER-BASED MEMORY OPTIMIZATION

The proposed model introduces hierarchical clustering combined with structured routing and memory optimization.

IV-A. Architecture Assumptions

- Each cluster contains **4 devices**: 1 Cluster Head (CH) and 3 Member Devices.
- The total number of clusters is $C = \frac{N}{4}$.
- CHs form a higher-level communication network.

IV-B. Message Flow

The message flow is structured as follows:

Source Device $\rightarrow$ Source Cluster Head $\rightarrow$ Cluster Head Network $\rightarrow$ Destination Cluster Head $\rightarrow$ Destination Device

IV-C. Communication Complexity (Transmission Analysis)

The proposed hierarchical clustering model reduces communication complexity by dividing the network into smaller clusters and using targeted routing.

1) Source device to source CH = 1 transmission.

2) Inter-cluster routing among CHs = $C-1$ transmissions.
3) Destination CH to destination device = 1 transmission.

The total number of transmissions is given by:

$$\boxed{T_{proposed} = 1 + (C-1) + 1 = C + 1 = \frac{N}{4} + 1} \tag{5}$$

Both models exhibit linear complexity $O(N)$, but the proposed model has a much smaller constant factor compared to the existing flooding model.

IV-D. Hop Count Complexity

Since routing is performed over a cluster-head hierarchy rather than unstructured flooding, the hop count grows logarithmically:

$$\boxed{H_{proposed} = O(\log N)} \tag{6}$$

IV-E. Memory Optimization

A Bloom filter stores message identifiers using a compact bit array rather than explicitly storing each identifier. When a message is received, multiple hash functions are applied to its identifier, and the corresponding bit positions are set in the array. To check for duplicates, the same hash functions are applied, and the corresponding bits are examined. If all bits are set, the message is considered a probable duplicate; otherwise, it is treated as new.

The memory requirement per element in a Bloom filter is given by:

$$\boxed{c = -\frac{\ln p}{(\ln 2)^2}} \tag{7}$$

where p represents the false positive probability. 72 byte per message UUID is reduced to approximately 1.2 bytes per element in bit array.

$$\boxed{c = -\frac{\ln p}{(\ln 2)^2} = 9.6 \text{ bits or } 1.2 \text{ byte}} \tag{8}$$

$$\boxed{M = N \times c \implies M = O(N)} \tag{9}$$

IV-F. Advantages

- Eliminates redundant rebroadcasting
- Reduces transmission overhead from $3N+1$ to $N/4+1$
- Reduces hop count from $O(N)$ to $O(\log N)$
- Lowers energy consumption
- Improves scalability
- Compatible with practical Nearby Connections peer limits

TABLE I
PERFORMANCE METRICS COMPARISON (BASE 10 MEMORY)

N	T_{Flood}	T_{Prop}	H_{Flood}	H_{Prop}	UUID (KB)	Bloom (KB)
100	301	26	25	4	7.20	0.12
200	601	51	50	5	14.40	0.24
300	901	76	75	5	21.60	0.36
400	1201	101	100	6	28.80	0.48
500	1501	126	125	6	36.00	0.60

V. RESULTS & COMPARATIVE ANALYSIS

The comparative analysis highlights significant improvements across all performance metrics.

V-A. Communication Improvements

- Controlled Flooding ($N = 1000$): 3001 transmissions
- Proposed Hierarchical Routing ($N = 1000$): 251 transmissions
- Improvement: $\approx 12\times$ reduction

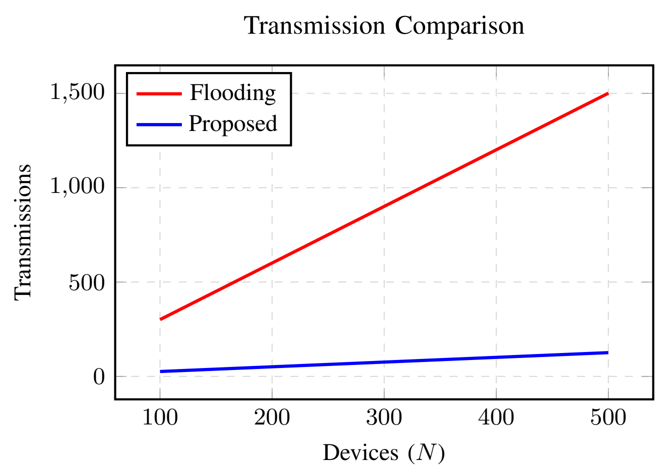

Fig. 2. Transmission Comparison: Comparison of transmission overhead.

V-B. Latency (Hop Count) Improvements

- Flooding: $O(N)$ growth
- Proposed: $O(\log N)$ reduction

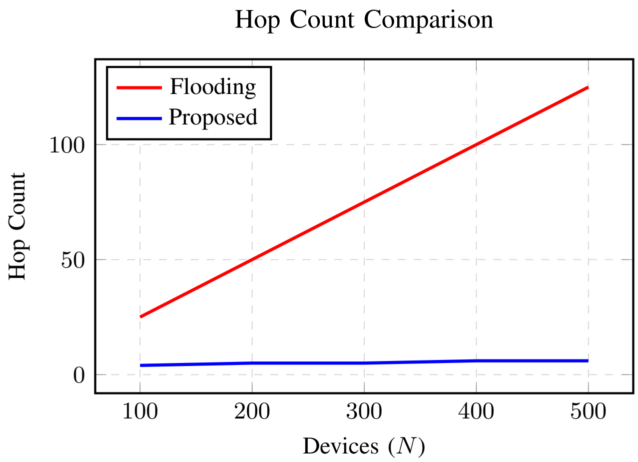

Fig. 3. Hop Count Comparison: Linear vs logarithmic reduction.

V-C. Memory Usage Improvements

- UUID Storage: 7.2 MB (100,000 messages)
- Bloom Filter: 120 KB
- Improvement: $\approx 60\times$ reduction

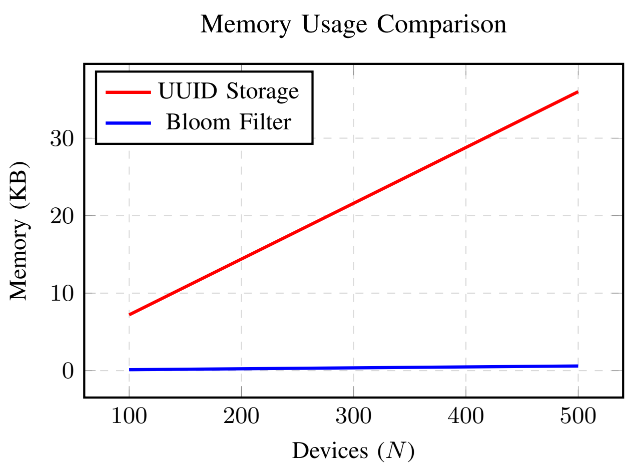

Fig. 4. Memory Usage Comparison: UUID vs Bloom Filter (KB).

V-D. Observations

- The current BluDrop system uses controlled flooding, while the proposed model uses hierarchical clustering and targeted routing.
- The transmission complexity remains $O(N)$, but with a significantly smaller constant factor.
- The routing decision complexity can be $O(\log N)$, which reduces hop count and latency.
- Memory optimization significantly reduces resource consumption.

VI. CONCLUSION

This paper analyzed the limitations of flooding-based peer-to-peer communication systems and proposed a hierarchical clustering-based routing model combined with Bloom filter-based memory optimization. The results demonstrate significant improvements in communication efficiency, latency reduction, and memory utilization.

By transforming unstructured communication into a structured and scalable architecture, the proposed model enables efficient operation in large-scale networks. Future work may include dynamic cluster formation, adaptive routing strategies, and real-world deployment for performance validation.

REFERENCES

[1] Broadcast Storms: S. Ni, Y. Tseng, Y. Chen, and J. Sheu, “The broadcast storm problem in a mobile ad hoc network,” *Proceedings of the 5th ACM/IEEE International Conference on Mobile Computing and Networking (MobiCom)*, 1999, pp. 151–162.

[2] AODV Routing: C. E. Perkins, E. M. Belding-Royer, and S. R. Das, “Ad hoc On-Demand Distance Vector (AODV) Routing,” *RFC 3561, IETF*, 2003.

[3] Clustering (HEED): O. Younis and S. Fahmy, “HEED: A hybrid, energy-efficient, distributed clustering approach for ad hoc sensor networks,” *IEEE Transactions on Mobile Computing*, vol. 3, no. 4, pp. 366–379, 2004.

[4] Google API: Google, “Nearby Connections API.”

[5] Bluetooth Standards: Bluetooth SIG, “Bluetooth Core Specification v5.0,” 2016.

[6] Bloom Filters: A. Broder and M. Mitzenmacher, “Network applications of Bloom filters: A survey,” *Internet Mathematics*, vol. 1, no. 4, pp. 485–509, 2004.

[7] Distributed Systems Theory: A. S. Tanenbaum and M. Van Steen, *Distributed Systems: Principles and Paradigms*, 2nd ed., Prentice Hall, 2007.